

Review retrospective:
*global constancy of the harmonic covariance, and a namespace
the reviewer mis-guessed*

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Abstract

A soundness review of `Laplace/OneD/HarmonicCovGlobalConstancy.lean`: the closed-form proof that the perturbed harmonic Gibbs covariance is globally constant, $\text{Cov}_h(x, x) = 1/(\lambda t)$ for all h . Result: **a clean fidelity pass with five edits** — a dead `open Real` and four dead proof-locals. The estimate is correct (the h -dependence cancels exactly), and the one disputed cleanup — whether `open Real` is dead — was settled by the build against the reviewer.

1 Setting

This is the file the cross-susceptibility capstone now defers to: where the κ_3 route gave only $\partial_h \text{Cov}_h(x, x)|_0 = 0$, this file proves the global statement by direct closed-form evaluation — square completion $(\lambda/2)x^2 + hx = (\lambda/2)(x + h/\lambda)^2 - h^2/(2\lambda)$ turns the perturbed Gibbs into a translate of the unperturbed Gaussian, and translation-invariance of Lebesgue (`integral_add_right_eq_self`) reduces every perturbed integral to an unperturbed moment.

2 What was reviewed

Nine private lemmas (square completion, the shift identity, three integrability witnesses, three unperturbed-moment closed forms) and four public theorems: the perturbed partition $Z_h = e^{th^2/2\lambda} \sqrt{2\pi/\lambda t}$, the centroid $\langle x \rangle_h = -h/\lambda$, the second moment $\langle x^2 \rangle_h = 1/(\lambda t) + h^2/\lambda^2$, and the headline covariance.

3 Fidelity / soundness findings

Sound; checked end to end. The reviewer and I independently confirmed all seven steps: the square-completion identity; the partition closed form (with the correct $e^{th^2/2\lambda}$ constant); the centroid $-h/\lambda$; the second moment as variance $1/(\lambda t)$ plus shifted-mean square $(h/\lambda)^2$; the unperturbed $\langle y^2 \rangle_0 = (1/(\lambda t))Z_0$; and — the headline — $\text{Cov}_h = \langle x^2 \rangle_h - \langle x \rangle_h^2 = (1/(\lambda t) + h^2/\lambda^2) - (h/\lambda)^2 = 1/(\lambda t)$, the h -dependence cancelling exactly. Docstrings accurate, including the “strengthens the κ_3 derivative-at-0 result to global” claim.

4 Simplifications applied

Five. A dead `open Real` (all `Real.*` qualified; `open MeasureTheory` kept for the 34 bare $\int$, `open scoped Nat` kept for the double-factorial `!!` at the $(2 \cdot 1 - 1)!!$ step), and four dead locals: `hZpos` (a $0 < \sqrt{\cdots}$ fact, never referenced — and, being a $0 <$ rather than $\neq 0$ fact, not `field_simp` fuel) and `hden_ne` (an $\int \cdots \neq 0$ fact that goes dead the moment `rw [hden]` replaces the denominator with its closed form, after which `hcommon_ne` is the witness `field_simp` uses), each in both the centroid and second-moment theorems. Build green at 8253 jobs, warning count 0, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

One reviewer finding rejected by the build. GPT claimed `open Real` was *not* dead, on the grounds that `integrable_exp_neg_mul_sq` and its two siblings are unqualified `Real`-namespace names. But removing `open Real` builds green — those lemmas are root-namespace Mathlib lemmas (reachable without `open Real`), so the open was genuinely dead. The build is the arbiter of a namespace claim, and it overruled the reviewer here, exactly as it has overruled *me* elsewhere (the `SemiDegenerate` transitive-import case).

6 Lessons

- **A $0 <$ fact is not `field_simp` fuel; a $\neq 0$ fact is — but only while its subject is in the goal.** `hZpos` ($0 < \sqrt{\cdots}$) was never fuel; `hden_ne` ($\int \neq 0$) *was* a $\neq 0$ fact, yet went dead once `rw [hden]` rewrote the $\int$ away, leaving `hcommon_ne` as the live witness. The discriminator for “is this $\neq 0$ local still load-bearing?” is whether its subject survives the rewrites before the `field_simp`.
- **The reviewer’s namespace guesses are build-checkable, both ways.** GPT was right about four dead locals and wrong about `open Real`; a single `lake build` adjudicated all five. Trusting the reviewer’s namespace recall over the green build would have re-introduced dead code.

7 Follow-ups

- Remaining `OneD/:` `HarmonicPerturbedMoments` (528L), `HarmonicGibbsObservableMonomials/HarmonicGibbsRegularity`, the `Anharmonic* arc`, and `IntegralRemainder` (2668L).
- Proof-golf forward tide: the `hrewrite/bridge-lemma simp` shortenings the reviewer flagged.